Sequences & Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Arithmetic sequences**

For the arithmetic sequence with $a_1 = 9$ and common difference $d = 6$, find a_{10}.

$a_{10} = a_1 + (10 - 1)d = 9 + 9(6) = 63.$

2 **[Foundational]** • **Target: Arithmetic sequences**

For the arithmetic sequence with $a_1 = 15$ and $d = -4$, find a_{11}.

$a_{11} = 15 + 10(-4) = 15 - 40 = -25.$

3 **[Foundational]** • **Target: Arithmetic series**

Find S_{10} for the arithmetic sequence with $a_1 = 9$, $d = 6$ (using $a_{10} = 63$ from above).

$S_{10} = \frac{n(a_1 + a_n)}{2} = \frac{10(9 + 63)}{2} = 5(72) = 360.$

4 **[Foundational]** • **Target: Geometric sequences**

For the geometric sequence with $a_1 = 6$ and common ratio $r = 2$, find a_8.

$a_8 = a_1 r^7 = 6(2^7) = 6(128) = 768.$

5 **[Foundational]** • **Target: Geometric sequences**

A geometric sequence has $a_1 = 2$ and common ratio $r = 5$. Find the smallest n for which $a_n = 1250$.

$a_n = 2 \cdot 5^{n-1}$. Setting $2 \cdot 5^{n-1} = 1250$ gives $5^{n-1} = 625 = 5^4$, so $n - 1 = 4$ and $n = 5$.

6 **[Foundational]** • **Target: Geometric series (finite)**

Find S_8 for the geometric sequence with $a_1 = 6$, $r = 2$ (using $a_8 = 768$ from above).

$S_8 = \frac{a_1(r^8 - 1)}{r - 1} = \frac{6(256 - 1)}{1} = 6(255) = 1530.$

7 **[Foundational]** • **Target: Infinite geometric series**

Find the sum of the infinite geometric series $36 + 12 + 4 + \cdots$.

$a = 36$, $r = \frac{1}{3}$, and $|r| < 1$, so the sum is $\frac{36}{1 - \frac{1}{3}} = \frac{36}{\frac{2}{3}} = 54.$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Geometric sequences

[Sequences & Series · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

9 [Intermediate] • Target: Geometric series (finite)

Express the repeating decimal $0.3\overline{18} = 0.3181818\ldots$ as an irreducible fraction using the infinite geometric series formula.

Step 1: Separate the non-repeating and repeating components: $0.3\overline{18} = \frac{3}{10} + 0.0181818\ldots$ Step 2: Represent the repeating part as an infinite geometric series: $0.0181818\ldots = \frac{18}{1000} + \frac{18}{100000} + \cdots$ Here the initial term is $a_1 = \frac{18}{1000}$ and the common ratio is $r = \frac{1}{100}$. Step 3: Sum the geometric series.

10 [Intermediate] • Target: Infinite geometric series

The second term of a geometric sequence is 12 and the fifth term is $\frac{32}{9}$. Find the common ratio r, the first term a_1, and the sum of the infinite series S_∞.

Step 1: Express terms in terms of a_1 and r: $a_2 = a_1 r = 12$ $a_5 = a_1 r^4 = \frac{32}{9}$ Step 2: Divide a_5 by a_2: $\frac{a_1 r^4}{a_1 r} = r^3 = \frac{32/9}{12} = \frac{32}{108} = \frac{8}{27}$ Step 3: Extract the common ratio r: $r = \sqrt[3]{\frac{8}{27}} = \mathbf{\frac{2}{3}}$. Step 4: Find a_1: $a_1\left(\frac{2}{3}\right) = 12 \Rightarrow a_1 = 12 \times \frac{3}{2} = \mathbf{18}$. Step 5: Compute the infinite sum S_∞: Since \(|r| = \frac{2}{3}

11 [Intermediate] • Target: Arithmetic sequences

An infinite geometric series has first term $a_1 = 18$ and sum to infinity $S_\infty = 24$. (a) Find the common ratio r. (b) Calculate the third term a_3.

(a) Find common ratio r: $S_\infty = \frac{a_1}{1-r} \Rightarrow 24 = \frac{18}{1-r} \Rightarrow 1 - r = \frac{18}{24} = \frac{3}{4} \Rightarrow r = 1 - \frac{3}{4} = \mathbf{\frac{1}{4}}$. (b) Calculate 3rd term a_3: $a_3 = a_1 \cdot r^2 = 18 \cdot \left(\frac{1}{4}\right)^2 = 18 \cdot \frac{1}{16} = \mathbf{\frac{9}{8}} \mathbf{= 1.125}$.

12 [Intermediate] • Target: Arithmetic series

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term a_1, and compute the exact value of the partial sum S_4.

Use the infinite sum formula $S_\infty = \frac{a_1}{1-r}$: $25 = \frac{a_1}{1-(-0.4)} = \frac{a_1}{1.4} \Rightarrow a_1 = 25 \times 1.4 = 35.$ Compute the 4th partial sum $S_4 = \frac{a_1(1-r^4)}{1-r}$: $(-0.4)^4 = 0.0256.$ $S_4 = \frac{35(1-0.0256)}{1.4} = \frac{35(0.9744)}{1.4} = 25 \times 0.9744 = 24.36.$

13 [Intermediate] • Target: Geometric sequences

Find the infinite sum of the geometric series $12 + 8 + \frac{16}{3} + \frac{32}{9} + \ldots$. Verify the condition for convergence.

Step 1: Determine first term and common ratio: First term $a_1 = 12$. Common ratio $r = \frac{8}{12} = \frac{2}{3}$. Step 2: Check convergence condition: Since $|r| = \left|\frac{2}{3}\right| = \frac{2}{3} < 1$, the infinite series converges. Step 3: Compute infinite sum: $S_\infty = \frac{a_1}{1-r} = \frac{12}{1-2/3} = \frac{12}{1/3} = 36$.

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14 [Intermediate] • Target: Geometric series (finite)

In an arithmetic sequence, the 4th term is 17 and the 10th term is 41. Determine the common difference d, the first term a_1, and the partial sum of terms from a_{11} through a_{25} inclusive.

Step 1: Solve for common difference d: $a_{10} - a_4 = (10-4)d = 6d$ $41 - 17 = 24 \Rightarrow 6d = 24 \Rightarrow \mathbf{d = 4}$. Step 2: Solve for the first term a_1: $a_4 = a_1 + 3d \Rightarrow 17 = a_1 + 3(4) \Rightarrow a_1 = 17 - 12 = \mathbf{5}$. Step 3: Find boundary terms for the range $n = 11$ to 25: Number of terms: $N = 25 - 11 + 1 = 15$. Term $a_{11} = a_1 + 10d = 5 + 10(4) = 45$. Term $a_{25} = a_1 + 24d = 5 + 24(4) = 5 + 96 = 101$. Step 4: Compute the sum.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 1.01 Complex Numbers

Evaluate the exact value of the telescoping sum: $\sum_{k=1}^{50} \left(\frac{1}{k} - \frac{1}{k+1}\right)$ and evaluate the convergent infinite series: $\sum_{n=1}^{\infty} \frac{3}{4^n}$.

Part 1: Telescoping sum: Write out the first few and last terms: $\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{50} - \frac{1}{51}\right)$ All intermediate terms cancel pairwise, leaving: $1 - \frac{1}{51} = \mathbf{\frac{50}{51}}$. Part 2: Infinite series: Factor the numerator and identify initial term and ratio: $\sum_{n=1}^{\infty} \frac{3}{4^n} = \frac{3}{4} + \frac{3}{16} + \frac{3}{64} + \cdots$ Initial term $a_1 = \frac{3}{4}$, ratio $r = \frac{1}{4}$..

16 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 3.01 Functions & Graphs

Evaluate the finite sum: $\sum_{k=1}^{6} 3(2)^{k-1}$

Step 1: Identify geometric series parameters: For $k = 1$, $a_1 = 3(2)^0 = 3$. The common ratio is $r = 2$ and the number of terms is $n = 6$. Step 2: Apply finite geometric sum formula:

$S_6 = \frac{a_1(1-r^6)}{1-r} = \frac{3(1-2^6)}{1-2} = \frac{3(1-64)}{-1} = \frac{3(-63)}{-1} = 3 \times 63 = \mathbf{189}$.

17 [Challenging] • Target: Arithmetic series • Cross-Lesson: 4.01 Matrices & Vectors

[Probability · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

18 [Challenging] • Target: Geometric sequences • Cross-Lesson: 5.01 Introductory Calculus

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.

In the binomial theorem expansion $\sum_{k=0}^{n} \binom{n}{k} A^{n-k} B^k$, the 1st term has $k = 0$, so the 6th term has $k = 5$. Identify terms: $n = 8$, $A = 3x$, $B = -2y^2$: $T_6 = \binom{8}{5}(3x)^{8-5}(-2y^2)^5 = \binom{8}{3}(3x)^3(-2y^2)^5$. Compute each factor:$\binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56 (3x)^3 = 27x^3 (-2y^2)^5 = -32y^{10}$Multiply the coefficients:

$T_6 = 56 \times 27 \times (-32) x^3 y^{10} = 56 \times (-864) x^3 y^{10} = -48384 x^3 y^{10}$.

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19 [Challenging] • Target: Geometric series (finite) • Cross-Lesson: 6.01 Probability

Evaluate the sum of the convergent infinite series: $24 - 12 + 6 - 3 + \cdots$

Step 1: Identify parameters: $a_1 = 24$, $r = \frac{-12}{24} = -\frac{1}{2}$. Since $|r| = \frac{1}{2} < 1$, the series converges. Step 2: Apply infinite sum formula: $S_\infty = \frac{a_1}{1-r} = \frac{24}{1-(-1/2)} = \frac{24}{3/2} = 24 \times \frac{2}{3} = \mathbf{16}$.

20 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 1.01 Complex Numbers

Evaluate the finite geometric sum: $\sum_{k=1}^{8} 3 \cdot 2^{k-1}$

Step 1: Identify components: Initial term $a_1 = 3(2^0) = 3$, ratio $r = 2$, number of terms $n = 8$. Step 2: Apply finite geometric sum formula: $S_8 = \frac{a_1(r^8-1)}{r-1} = \frac{3(2^8-1)}{2-1} = 3(256-1) = 3(255) = \mathbf{765}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Arithmetic sequences

Find S_6 for the arithmetic sequence with $a_1 = -13$, $d = -1$.

a. -94

b. -89

c. -97

d. -93 ✓

$a_6 = -13 + (5)(-1) = -18$. $S_6 = \frac{6(-13-18)}{2} = 3(-31) = -93$.

22 [Foundational] • Target: Arithmetic series

Find S_5 for the geometric sequence with $a_1 = -3$, $r = 3$.

a. -361

b. -363 ✓

c. -366

d. -365

$S_5 = \frac{a_1(3^5-1)}{r-1} = \frac{-3(242)}{2} = -363$.

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23 [Foundational] • Target: Geometric sequences

Does the infinite geometric series with $a_1 = 6$ and $r = 1$ converge?

a. No — $|r| \geq 1$, it diverges. ✓

b. Yes — it converges to $\frac{a_1}{1-r}$.

c. Yes — every geometric series converges.

d. Cannot be determined.

$|r| = 1 \geq 1$, so by Theorem 3.2 the series diverges (has no finite sum).

24 [Foundational] • Target: Geometric series (finite)

Consider a geometric sequence where the first term is $a_1 = \frac{1}{2}$ and the common ratio is r. If the sum of the first three terms is $S_3 = \frac{7}{2}$, find the value of the common ratio r.

a. 2 ✓

b. -2

c. 1/2

d. 3

The sum of the first three terms of a geometric sequence is given by the formula $S_n = a_1\frac{1-r^n}{1-r}$ for $req1$. We are given $a_1 = \frac{1}{2}$ and $S_3 = \frac{7}{2}$. Substituting these values into the formula: $\frac{7}{2} = \frac{1}{2} \cdot \frac{1-r^3}{1-r}$ Multiply both sides by 2: $7 = \frac{1-r^3}{1-r}$ We know that $1 - r^3 = (1 - r)(1 + r + r^2)$. So, for $req1$, we can simplify the expression: $7 = 1 + r + r^2$ Rearrange into a quadratic equation: $r^2 + r + 1 - 7 = 0$ $r^2 + r - 6 = 0$ Factor the.

25 [Foundational] • Target: Infinite geometric series

A geometric sequence has its first term a_1 and a common ratio r. The sum of the first three terms is 26, and the sum of the first six terms is 728. If $a_1 = 2$, find the value of r.

a. 2

b. 3 ✓

c. -2

d. -3

Let the first term of the geometric sequence be a_1 and the common ratio be r. The sum of the first n terms of a geometric sequence is given by $S_n = a_1\frac{1-r^n}{1-r}$ (for $req1$). We are given that $a_1 = 2$. The sum of the first three terms is $S_3 = 26$. So, $2\frac{1-r^3}{1-r} = 26$. This simplifies to $\frac{1-r^3}{1-r} = 13$. Using the difference of cubes formula, $1 - r^3 = (1 - r)(1 + r + r^2)$. Thus, $1 + r + r^2 = 13$. Rearranging this, we get a quadratic equation.

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26 [Foundational] • Target: Arithmetic sequences

A sequence is defined by the recurrence relation $a_{n+1} = 2a_n - 3$ for $n \geq 1$, with $a_1 = 5$. Consider a second sequence, b_n, where $b_n = a_n - k$ for some constant k. If the sequence b_n is a geometric sequence, find the value of k and the common ratio of the sequence b_n.

a. k = 3, r = 2 ✓

b. k = -3, r = 2

c. k = 2, r = 3

d. k = 3, r = 3

The first sequence is defined by $a_{n+1} = 2a_n - 3$, with $a_1 = 5$. We are given a second sequence $b_n = a_n - k$. For b_n to be a geometric sequence, the ratio $\frac{b_{n+1}}{b_n}$ must be constant. Substituting $a_n = b_n + k$ into the recurrence relation for a_n: $b_{n+1} + k = 2(b_n + k) - 3$ $b_{n+1} + k = 2b_n + 2k - 3$ $b_{n+1} = 2b_n + k - 3$ For b_n to be a geometric sequence with common ratio r, we must have $b_{n+1} = rb_n$. Comparing this with the derived equation, we see that r must be.

27 [Foundational] • Target: Arithmetic series

A geometric sequence has $a_1 = 6$ and common ratio $r = 2$. Find a_5.

a. 96 ✓

b. 192

c. 98

d. -96

Applying Sequences & Series principles, $a_5 = a_1 r^{n-1} = 6(2)^4 = 96$. Therefore, the verified correct option is 96.

28 [Foundational] • Target: Geometric sequences

Find the common ratio of the geometric sequence $5,\ 10,\ 20,\ 40,\ \ldots$

a. 2 ✓

b. -2

c. 3

d. 5

Divide consecutive terms: $10 \div 5 = 2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Geometric series (finite)

Find the sum of the first 6 terms of the arithmetic sequence with $a_1 = 9$ and $d = 3$.

a. **99** ✓

b. 105

c. 144

d. 96

$S_6 = \frac{6}{2}(2(9) + (5)(3)) = 99$.

30 [Intermediate] • Target: Infinite geometric series

An arithmetic sequence (a_n) and a geometric sequence (b_n) of non-zero real numbers share the same first term, so $a_1 = b_1$. The sequences also satisfy $a_5 = b_2$ and $a_6 = b_3$, with $b_2 eq b_1$. If the maximum value of the partial sum $S_N = \sum_{k=1}^{N} a_k$ of the arithmetic sequence is 153, what is the sum of the infinite geometric series $\sum_{n=1}^{\infty} b_n$?

a. **64** ✓

b. 96

c. 144

d. 192

Let $a_1 = b_1 = a$, let d be the common difference of (a_n), and let r be the common ratio of (b_n). 1. Express the given terms: $a_5 = a + 4d = b_2 = ar$ $a_6 = a + 5d = b_3 = ar^2$ Subtracting the first equation from the second gives: $d = a_6 - a_5 = ar^2 - ar = ar(r - 1)$ Substitute d back into $a + 4d = ar$:
$a + 4ar(r - 1) = ar$ Since the terms of the sequence are non-zero, $a eq 0$. Dividing by a:
$1 + 4r(r - 1) = r \Rightarrow 4r^2 - 5r + 1 = 0$ $(4r - 1)(r - 1) = 0$ Since $b_2 eq b_1$.

31 [Intermediate] • Target: Arithmetic sequences

The graph of a linear function $f(x)$ is shown below. What is the value of $f(0)$?

a. -2

b. 0

c. 2 ✓

d. 4

The value of $f(0)$ represents the y-intercept of the graph. On the graph, the line crosses the y-axis at the point $(0, 2)$. Therefore, $f(0) = 2$.

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32 [Intermediate] • Target: Arithmetic series

An infinite geometric sequence $a_1, a_2, a_3, \ldots$ of positive real numbers has common ratio $r < 1$ and sum S. The sub-sequence of terms whose indices are of the form $3k-1$ for positive integers k (that is, $a_2, a_5, a_8, \ldots$) has sum $\sum_{}^{\infty} a_{3k-1} = \frac{3}{13}S$. Furthermore, the infinite series formed by the squares of the terms has sum $\sum_{n=1}^{\infty} a_n^2 = \frac{9}{2}S$. What is the value of $\sum_{n=1}^{\infty} na_n$?

a. $\frac{27}{2}$ ✓

b. $\frac{81}{4}$

c. $\frac{9}{2}$

d. $\frac{27}{4}$

Let the first term be $a_1 = a > 0$ and the common ratio be $r \in (0, 1)$. The sum of the original series is: $S = \sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$. **Step 1: Determine the common ratio r.** The sub-sequence consists of terms $a_2, a_5, a_8, \ldots$, which has first term $a_2 = ar$ and common ratio r^3. Its sum is: $\sum_{k=1}^{\infty} a_{3k-1} = \frac{ar}{1-r^3}$. Given that this sum is $\frac{3}{13}S$: $\frac{ar}{1-r^3} = \frac{3}{13} \cdot \frac{a}{1-r}$. Since $a > 0$ and.

33 [Intermediate] • Target: Geometric sequences

An infinite geometric sequence of positive real numbers has first term a_1 and common ratio $r \in (0, 1)$. Let $S_1 = \sum_{n=1}^{\infty} a_n$, $S_2 = \sum_{n=1}^{\infty} a_n^2$, and $S_3 = \sum_{n=1}^{\infty} a_n^3$. If $S_1 = 4$ and $S_2 = 8$, what is the value of S_3?

a. $\frac{256}{13}$ ✓

b. $\frac{128}{13}$

c. $\frac{64}{7}$

d. 16

For an infinite geometric sequence with first term $a_1 > 0$ and ratio $r \in (0, 1)$, the terms are $a_n = a_1 r^{n-1}$. 1. The sum of the original sequence is: $S_1 = \frac{a_1}{1-r} = 4 \Rightarrow a_1 = 4(1-r)$ 2. The terms of the squared sequence are $a_n^2 = a_1^2 (r^2)^{n-1}$, which forms a geometric sequence with first term a_1^2 and common ratio r^2. Its sum is: $S_2 = \frac{a_1^2}{1-r^2} = 8$ 3. Relate S_2 and S_1 using the algebraic identity $1 - r^2 = (1-r)(1+r)$.

34 [Intermediate] • Target: Geometric series (finite)

The graph of the function $y = f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Determine the value of b.

a. **0** ✓

b. -2

c. 2

d. 1

The general form of the quadratic function is $y = ax^2 + bx + c$. We are given three points that lie on the graph: $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Substituting these points into the equation, we get a system of three linear equations with three unknowns (a, b, and c): 1. For $(-1, 5)$: $a(-1)^2 + b(-1) + c = 5 \Rightarrow a - b + c = 5$ 2. For $(1, 1)$: $a(1)^2 + b(1) + c = 1 \Rightarrow a + b + c = 1$ 3. For $(3, 5)$: $a(3)^2 + b(3) + c = 5 \Rightarrow 9a + 3b + c = 5$ We can solve this system of equations. Le.

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35 [Intermediate] • Target: Infinite geometric series

Consider the function $f(x) = x^3 - 6x^2 + 9x$. Find the area of the region bounded by the graph of $f(x)$ and the x-axis.

a. 10

b. 27/2

c. 9

d. **27/4 ✓**

First, find the x-intercepts of the function by setting $f(x) = 0$: $x^3 - 6x^2 + 9x = 0$ $x(x^2 - 6x + 9) = 0$ $x(x - 3)^2 = 0$ The x-intercepts are $x = 0$ and $x = 3$ (a repeated root, indicating the graph touches the x-axis at $x = 3$ without crossing). To find the area, we need to integrate the function between these intercepts. We need to determine if the function is above or below the x-axis in the interval $(0, 3)$. Let's test a point, say $x = 1$: $f(1) = 1^3 - 6(1)^2 + 9(1) = 1 - 6 + 9 = 4$. Since $f(1) > 0$.

36 [Intermediate] • Target: Arithmetic sequences

The first term of a geometric sequence is a. The common ratio is r. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. If $r > 0$, find the value of a and r.

a. $a = 3, r = 2$ ✓

b. $a = 7, r = 1/2$

c. $a = 3, r = \sqrt{2}$

d. $a = 21, r = 1$

Let the geometric sequence be denoted by a_n, where $a_1 = a$ and the common ratio is r. The sum of the first n terms of a geometric sequence is given by $S_n = \frac{a(r^n - 1)}{r - 1}$ (for $req1$). We are given that $S_3 = 21$ and $S_6 = 189$. So, we have the equations: 1) $\frac{a(r^3 - 1)}{r - 1} = 21$ 2) $\frac{a(r^6 - 1)}{r - 1} = 189$ We can rewrite $r^6 - 1$ using the difference of squares factorization: $r^6 - 1 = (r^3)^2 - 1^2 = (r^3 - 1)(r^3 + 1)$. Substitute this into the second equation.

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37 [Intermediate] • Target: Arithmetic series

The first three terms of a geometric sequence of positive real numbers are $k-1$, $k+2$, and $3k$, in that order. What is the value of the 6th term of this sequence?

a. 24

b. 48

c. **96** ✓

d. 192

Step 1: Set up the ratio property for consecutive terms of a geometric sequence. Since the sequence is geometric, the common ratio r satisfies: $r = \frac{a_2}{a_1} = \frac{a_3}{a_2} \Rightarrow \frac{k+2}{k-1} = \frac{3k}{k+2}$ Step 2: Solve the quadratic equation for k. Cross-multiplying gives: $(k+2)^2 = 3k(k-1)$ $k^2 + 4k + 4 = 3k^2 - 3k$ $2k^2 - 7k - 4 = 0$ Factoring the quadratic: $(2k+1)(k-4) = 0$ Thus, $k = -\frac{1}{2}$ or $k = 4$. Step 3: Discard the extraneous solution. The problem stat.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Geometric sequences • Cross-Lesson: 5.01 Introductory Calculus

The second term of a geometric sequence of positive real numbers is k, and the fifth term is $k^2 - 20$. If the common ratio of the sequence is 2, what is the sum of the first 6 terms of the sequence?

a. 155

b. 160

c. **315** ✓

d. 630

Step 1: Express the terms in terms of the first term a_1 and common ratio $r = 2$. The n-th term of a geometric sequence is given by $a_n = a_1 r^{n-1}$. For the second term: $a_2 = a_1(2)^{2-1} = 2a_1 = k \Rightarrow a_1 = \frac{k}{2}$ For the fifth term: $a_5 = a_1(2)^{5-1} = 16a_1$ Substituting $a_1 = \frac{k}{2}$ gives: $a_5 = 16\left(\frac{k}{2}\right) = 8k$ Step 2: Solve for the parameter k. We are given $a_5 = k^2 - 20$, so: $k^2 - 20 = 8k \Rightarrow k^2 - 8k - 20 = 0$ Factoring the quadratic equation.

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39 [Challenging] • Target: Geometric series (finite) • Cross-Lesson: 6.01 Probability

Consider the arithmetic sequence with first term $a_1 = 5$ and a common difference $d = 3$. What is the formula for the n-th term, a_n?

a. $a_n = 5 + 3(n - 1)$ ✓

b. $a_n = 3n + 2$

c. $a_n = 5n + 3$

d. $a_n = 5 + 3n$

The formula for the n-th term of an arithmetic sequence is given by $a_n = a_1 + d(n - 1)$, where a_1 is the first term and d is the common difference. In this case, $a_1 = 5$ and $d = 3$. Substituting these values into the formula gives $a_n = 5 + 3(n - 1)$. If we simplify this, we get $a_n = 5 + 3n - 3 = 3n + 2$. Therefore, the direct formula is $a_n = 5 + 3(n - 1)$ or its simplified form $a_n = 3n + 2$. Choice A directly uses the definition. Choice B is the simplified form. Let's examine all options: Choice A.

40 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 1.01 Complex Numbers

An arithmetic sequence has its n-th term given by $a_n = 3n + k$, where k is a constant. If the sum of the first 5 terms of this sequence is 65, find the value of k.

a. 1

b. **2** ✓

c. 3

d. 4

The sum of the first n terms of an arithmetic sequence is given by $S_n = \frac{n}{2}(2a_1 + (n - 1)d)$ or $S_n = \frac{n}{2}(a_1 + a_n)$. In this case, $a_n = 3n + k$. So, $a_1 = 3(1) + k = 3 + k$. The common difference is $d = a_{n+1} - a_n = (3(n + 1) + k) - (3n + k) = 3n + 3 + k - 3n - k = 3$. The sum of the first 5 terms is $S_5 = 65$. Using the formula $S_n = \frac{n}{2}(a_1 + a_n)$, we have $S_5 = \frac{5}{2}(a_1 + a_5)$. We need $a_5 = 3(5) + k = 15 + k$. Therefore, $65 = \frac{5}{2}((3 + k) + (15 + k))$. Multiply both sides b.

41 [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 3.01 Functions & Graphs

The graph of the function $f(x) = x^2 - 6x + k$ is shown below. The graph has a vertex at $(3, -5)$. Determine the value of the parameter k.

a. 4

b. **1** ✓

c. 9

d. -4

The vertex of a parabola in the form $f(x) = ax^2 + bx + c$ occurs at $x = -b/(2a)$. For $f(x) = x^2 - 6x + k$, we have $a = 1$ and $b = -6$. The x-coordinate of the vertex is $-(-6)/(2 \cdot 1) = 6/2 = 3$, which matches the given vertex. The y-coordinate of the vertex is $f(3)$. We are given that the vertex is $(3, -5)$, so $f(3) = -5$. Substituting $x = 3$ into the function: $f(3) = (3)^2 - 6(3) + k = 9 - 18 + k = -9 + k$. Setting this equal to the given y-coordinate of the vertex: $-9 + k = -5$. Solving for k.

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42 [Challenging] • Target: Arithmetic series • Cross-Lesson: 4.01 Matrices & Vectors

In an arithmetic sequence, the 3rd term is $k + 7$, the 6th term is $3k - 1$, and the 11th term is $6k - 8$, where k is a real constant. What is the first term, a_1, of this sequence?

a. 6 ✓

b. 10

c. 19

d. 26

In an arithmetic sequence with first term a_1 and common difference d, the nth term is given by $a_n = a_1 + (n - 1)d$. Step 1: Express the common difference d in terms of k using the differences between the given terms: $a_6 - a_3 = 3d = (3k - 1) - (k + 7) = 2k - 8 \Rightarrow d = \frac{2k-8}{3}$ $a_{11} - a_6 = 5d = (6k - 8) - (3k - 1) = 3k - 7 \Rightarrow d = \frac{3k-7}{5}$ Step 2: Equate the two expressions for d to solve for k: $\frac{2k-8}{3} = \frac{3k-7}{5}$ $5(2k - 8) = 3(3k - 7)$.

43 [Challenging] • Target: Geometric sequences • Cross-Lesson: 5.01 Introductory Calculus

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

a. 4

b. **5** ✓

c. 6

d. 7

The sum of the first n terms of a geometric sequence is given by the formula $S_n = a_1\frac{r^n-1}{r-1}$, where a_1 is the first term and r is the common ratio. Given $a_1 = 5$ and $r = 2$, and $S_n = 315$, we can substitute these values into the formula: $315 = 5\frac{2^n-1}{2-1}$ $315 = 5\frac{2^n-1}{1}$ $315 = 5(2^n - 1)$ Divide both sides by 5: $\frac{315}{5} = 2^n - 1$ $63 = 2^n - 1$ Add 1 to both sides: $63 + 1 = 2^n$ $64 = 2^n$ Since $2^6 = 64$, we have $n = 6$. Let's check the choices.

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44 [Challenging] • Target: Geometric series (finite) • Cross-Lesson: 6.01 Probability

Consider the functions $f(x) = x^2 - 6x + 5$ and $g(x) = -x + 5$. The graphs of these two functions intersect at two points. Let these points be A and B. A third function, $h(x)$, is defined as the difference between the x-coordinates of these intersection points, i.e., $h(x) = |x_A - x_B|$. What is the value of $h(x)$?

a. $\sqrt{16}$

b. $\sqrt{25}$ ✓

c. $\sqrt{36}$

d. $\sqrt{49}$

To find the intersection points of the graphs of $f(x)$ and $g(x)$, we set $f(x) = g(x)$. $x^2 - 6x + 5 = -x + 5$ Subtract $-x + 5$ from both sides: $x^2 - 6x + 5 - (-x + 5) = 0$ $x^2 - 6x + 5 + x - 5 = 0$ Combine like terms: $x^2 - 5x = 0$ Factor out x: $x(x - 5) = 0$ This gives two possible x-values for the intersection points: $x_1 = 0$ $x_2 = 5$ Let $x_A = 0$ and $x_B = 5$ (or vice versa). The problem defines $h(x)$ as the absolute difference between these x-coordinates: $h(x) = |x_A - x_B|$..

45 [Challenging] • Target: Infinite geometric series • Cross-Lesson: 1.01 Complex Numbers

In an arithmetic sequence, the first term is $a_1 = 4$ and the fifth term is $a_5 = 28$. What is the value of the tenth term, a_{10}?

a. 52

b. 56

c. **58** ✓

d. 64

The nth term of an arithmetic sequence is given by the formula: $a_n = a_1 + (n - 1)d$ where a_1 is the first term and d is the common difference. Using the fifth term $a_5 = 28$:
$28 = 4 + (5 - 1)d \Rightarrow 28 = 4 + 4d \Rightarrow 4d = 24 \Rightarrow d = 6$ Now, calculate the tenth term a_{10}:
$a_{10} = a_1 + (10 - 1)d = 4 + 9(6) = 4 + 54 = 58$ Common misconceptions: - Choosing 56 incorrectly assumes direct proportionality (e.g., $a_{10} = 2 \times a_5 = 2 \times 28$), which does not hold for arithmeti.

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